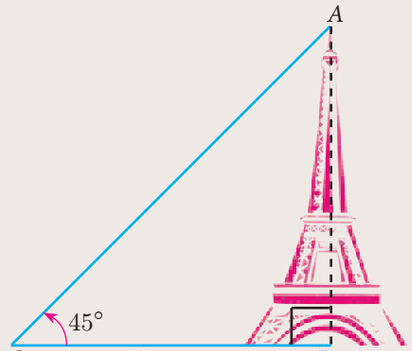
Activity 2

Representation of situations through right triangles. Draw a figure to illustrate the situation.

Situations	Draw a figure
A tower stands vertically on the ground. From a point on the ground, which is 20m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 45°.	 Fig. 6.11
An observer of 1.8 m tall is 25.2 m away from a chimney. The angle of elevation of the top of the chimney from her eyes is 45°.	……
From a point P on the ground the angle of elevation of the top of a 20 m tall building is 30°. A flag is hoisted at the top of the building and the angle of elevation of the top of the flagstaff from P is 55° .	……
The shadow of a tower standing on a level ground is found to be 40 m longer when the Sun's altitude is 30° than when it is 60°.	……

6.3.1 Problems involving Angle of Elevation

In this section, we try to solve problems when Angle of elevation are given.

Example 6.18

Calculate $\angle BAC$ in the given triangles. (tan 38.7° = 0.8011, tan 69.4° = 2.6604)

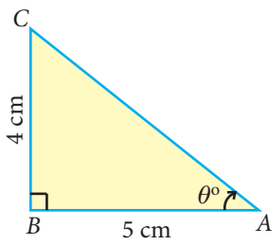


Fig. 6.12(a)

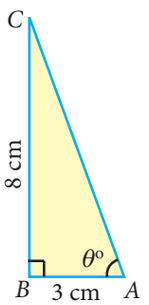


Fig. 6.12(b)

Solution

(i) In the right angled ΔABC [see Fig.6.12(a)]

$$\tan\theta = \frac{\text{opposite side}}{\text{adjacent side}} = \frac{4}{5}$$

$$\tan\theta = 0.8$$

$$\Rightarrow \theta = 38.7^\circ \ (\because \tan 38.7^\circ = 0.8011)$$

$$\therefore \angle BAC = 38.7^\circ$$

(ii) In the right angled ΔABC [see Fig.6.12(b)]

$$\tan\theta = \frac{8}{3}$$

$$\tan\theta = 2.66$$

$$\Rightarrow \theta = 69.4^\circ \ (\because \tan 69.4^\circ = 2.6604)$$

$$\therefore \angle BAC = 69.4^\circ$$

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